Bought, Not Added: Provisional Emission with Retrospective Revision for Streaming 3D Perception

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Abstract

Streaming 3D perception must emit objects immediately, yet its consumers need one physical object to stay one object. Temporal confirmation, which reports a track only after N observations, is the standard remedy, but it either delays those observations or discards them. At matched detection and identity budgets, and against a confidence-accumulation lifecycle, we find that its association gain is real but is paid for in detection recall, DetA and HOTA. The gain is bought rather than added. We therefore propose provisional emission with retrospective revision*: every observation is emitted at its own frame under a provisional status, and later confirmation revises the stream already delivered instead of withholding it. A four-frame retraction horizon exactly reproduces the unbounded retrospective output. On nuScenes it retains* 160,938 *observations (*+43.7%*) that prefix-deleting confirmation discards, at zero emission latency, and revision raises AMOTA by* 0.0823 *— while giving up DetA, HOTA and IDF1 against that arm. On ScanNet200 the same rule reduces class-label switching by* 26.9% *at a matched identity count. We offer a different operating point in the detection–association–latency–revisability trade-off, not a uniformly better tracker.*

1. Introduction

Streaming 3D perception commits online: a map or a planner's object memory consumes each frame as it arrives [10–12, 16]. Per-frame emission makes that commitment cheap and unstable — a re-detected object arrives as a new identity, an open-vocabulary label changes class between frames — and per-frame detection metrics neither see nor penalise it. Temporal confirmation is the natural correction, and an old one [3, 4, 32, 33]: report a track only once it has been observed N times, so transient errors never reach the consumer. But a rule that reports later necessarily reports less; it either delays commitment or withholds what it is not yet sure of, and both are paid on the streaming path.

Whether that buys anything is harder to establish than it looks. Confirmation does not merely reorder a stream — it changes how many detections enter it and how many identities the system carries — and detection- and association-oriented metrics are strongly coupled to both. A confirmed stream compared against an immediately emitted or score-thresholded one therefore differs in two things at once: a method emitting fewer instances, or maintaining fewer identities, can post a better association score without associating better. The benefit has to be measured at a *matched budget*, against controls granted the same number of emitted detections and, separately, the same number of emitted identities. This paper is not another confirmation rule; it asks whether the consistency benefit attributed to confirmation survives that comparison, and what it costs when it does.

It survives. On nuScenes with a frozen detector, confirmation improves association consistency under both matchings, in both association frames, and against confidence accumulation [1] as well as a static threshold, so the gain is not an artifact of carrying fewer identities. At the identical detection budget the same stream loses detection recall, DetA and aggregate HOTA. The gain is bought rather than added. Applied unchanged to an indoor open-vocabulary system sharing no detector or modality, the same rule reduces class-label switching by 26.9% at a matched identity count.

Both costs descend from one decision, that an observation is emitted only once its track is confirmed. Releasing the withheld observations after the fact delays them; discarding them instead reverses the association gain outright in the world frame. We therefore split the two decisions that rule fuses: *when an output is emitted* and *when its status is confirmed*. Every observation is emitted at the frame it occurs, marked provisional; later evidence revises the stream already delivered rather than withholding it, and a track that never confirms is retracted within a bounded window. Confirmation leaves the critical streaming path: the policy retains 160,938 observations (+43.7%) that prefix-deleting confirmation discards, at zero emission latency, and revision raises AMOTA by 0.0823. It is not free either: against that arm it gives up class-agnostic detection and identity quality.

Contributions.

- **A matched-budget diagnosis of temporal selection.** We identify the budget confound in comparisons of temporal emission rules and evaluate confirmation under both matchings. The design is not specific to confirmation and applies to any component that alters output volume.
- **The detection–association trade-off, characterised.** The association gain is accompanied by a retention cost: confirmation occupies a point on a trade-off rather than improving tracking uniformly.
- **Provisional emission with retrospective revision.** A causal policy decoupling output availability from evidence maturity, evaluated against the prefix-deleting confirmation arm it replaces.

2. Related Work

Streaming perception. Streaming perception scores predictions against the world at the moment they are *emitted*, folding latency into streaming AP (sAP) [16]; later work optimises detectors for this regime [35] and processes LiDAR sectors as they arrive rather than waiting for a full sweep [11]. sAP remains a per-frame average precision, so two systems emitting identical boxes at identical latency but differing entirely in identity continuity and label stability receive the same score.

Track lifecycle and confirmation. The classical radar literature decides when a candidate track is real enough to report with *track confirmation* — under M-out-of-N history logic a track is confirmed once it has been assigned a detection in M of the last N updates — and designs confirmation, deletion, and track management together [4]; the same literature established that reporting may lag estimation, *retrodiction* revising earlier estimates once later measurements arrive [15]. Tracking-by-detection inherited the count-based half: SORT [3] and DeepSORT [33] withhold a tentative track until matched in N consecutive frames and AB3DMOT [32] carries the same minimum-hits rule onto 3D boxes, while CenterPoint [36] reports tracks from their first observation. We use this confirmation test at its conventional operating point rather than proposing a new one.

Confidence in the track lifecycle. Recent 3D tracking has largely moved this decision from counting to confidence: CBMOT [1] replaces the minimum-hits criterion with a propagated tracklet score fused with each matched detection score, SimpleTrack [22] names *output* as a lifecycle decision separate from birth and death, and Poly-MOT [19] and Fast-Poly [20] pair category-specific association with mixed count- and confidence-based termination. We take this line as a comparator rather than a competitor: our associator is deliberately lightweight, so our absolute tracking scores are not comparable to theirs. Both rules decide which detections reach the output and both emit less, which is why we compare them at matched output and identity budgets.

Open-vocabulary 3D perception. This setting has moved from offline labelling of a completed reconstruction [5, 13, 23, 27] to incremental object maps [10, 12] and to fully online instance segmentation that maintains persistent identities across frames [28, 29, 34]. Cross-frame identity is thus an acknowledged design goal, but this work is scored by per-scene instance-segmentation AP, so how often a persistent instance's *class label* changes as observations accumulate is never quantified.

Evaluation metrics. CLEAR MOTA [2] and IDF1 [24] weight detection and identity errors differently, while HOTA [21] factorises the two into a detection accuracy DetA and an association accuracy AssA, which is what lets us read detection and association quality separately over an *identical* box set. In 3D, nuScenes AMOTA [6], following AB3DMOT's integral formulation [32], is class-aware and responds to labels the class-agnostic family ignores, whereas video-consistency and open-vocabulary tracking metrics (STQ [31], VPQ [14], TETA [17], TAO [9], OVTrack [18]) require dense ground-truth tube identities over a fixed taxonomy; we use existing metrics where computable.

Existing work targets streaming availability or instance-level consistency, but does not isolate temporal selection under matched output budgets, nor measure the class-label stability of persistent instances.

3. Method

3.1. Temporal lifecycle

Everything in this section is a post-processing operator over *frozen* detector output: no box coordinate or score is modified and no model is retrained, and the operator's only freedoms are which detections enter the output stream, under which persistent identity, with which label, and at which frame. Per-frame detections — a 3D box with a score and a class label — are linked across frames into tracks by a class-agnostic nearest-centroid associator with a $2.0\,\mathrm{m}$ bird's-eye-view **association threshold** and a **gap tolerance** of 5 frames; no score threshold is applied before association, and the associator is identical in every arm of every experiment. Association runs either in the *sensor frame* (raw detector coordinates) or in the *world frame* (ego-motion-compensated global coordinates), and both are evaluated throughout, because ego-motion compensation changes how hard association is. A track is **confirmed** once it has accumulated N observations; we use $N{=}3$ throughout, the conventional setting in tracking-by-detection systems, and do not tune it per dataset or per metric. Counting is *cumulative* over the scene rather than over consecutive frames, so a track seen

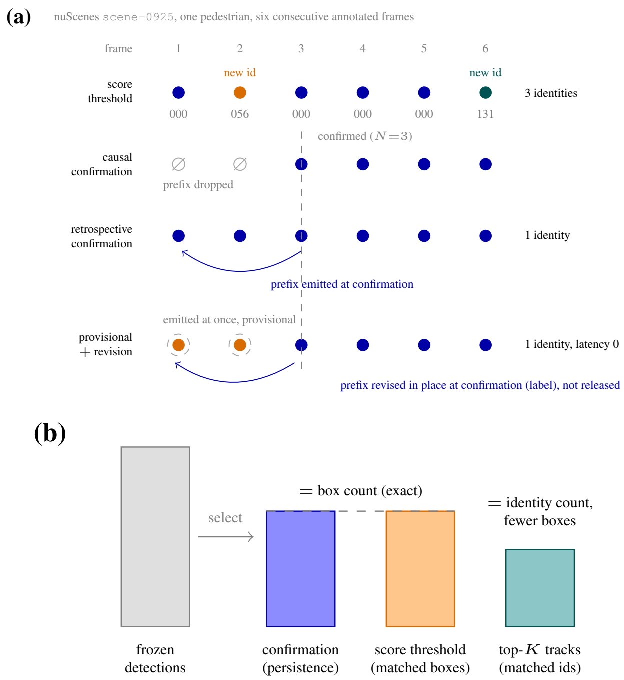

Figure 1. **Emission policies over the same frozen detections.** **(a)** Six consecutive annotated frames of nuScenes `scene-0925` under a score threshold, causal confirmation, retrospective confirmation, and our provisional emission with revision. Confirmation requires N=3; the threshold carries no history and assigns three identities (last three digits shown), while ours emits each observation immediately (dashed ring) and revises its label after confirmation. **(b)** The detection- and identity-budget controls match emitted boxes and emitted tracks respectively (Sec. 4.2).

at frames $0, 3, 6$ confirms at frame 6. An unconfirmed track is **retracted** once $t - t_{\text{last seen}} > H$, where H is the *retraction horizon* in frames; confirmed tracks are never retracted, and H=∞ applies no death rule and defers to a scene-end decision, which serves as the reference endpoint.

3.2. Emission policies

Over an identical lifecycle, three policies differ only in what the consumer sees and when (Fig. 1a). **Causal confirmation** emits nothing until confirmation and drops the pre-confirmation observations permanently. **Retrospective confirmation** emits nothing until confirmation, at which point the withheld prefix is released alongside the current observation, completing the trajectory at the price of emission latency. **Provisional emission with revision** (ours) emits every observation at its own frame and revises it once the evidence completes.

3.3. Provisional emission and revision

Every observation is emitted at the frame it occurs under a *provisional status*: a flag on an emitted observation recording that the evidence supporting it is not yet complete. It is a property of an emitted record, not of a withheld one. Confirmation issues no prefix, because none was withheld; it issues a *revision* over the observations already delivered. In our instantiation the revision rewrites the class label of the track's already-emitted observations and the nuScenes attribute derived from it; box geometry, detection score, and the persistent identity are not rewritten, so the set of emitted boxes is unchanged by a revision. A track that reaches the retraction horizon unconfirmed is retracted.

Retrospective confirmation incurs $N-1$ observations of latency, and a larger empirical frame-index delay when observations contain gaps (Sec. 5.3); our policy has zero emission latency — asserted mechanically, since every emitted record carries the frame index of its own observation — while H bounds retraction rather than emission. What the policy requires instead is a consumer that can act on an update to a record it has already received.

4. Evaluation Protocol

Every claim in this paper is made against a *matched control* — a non-temporal selection rule granted the same output budget — rather than against the unfiltered baseline.

4.1. Datasets and pipelines

Outdoor. The nuScenes [7] validation split, all 150 scenes at the 2 Hz annotated-frame rate. Detections come from a pretrained CenterPoint [36] detector whose per-frame output is computed once and *frozen*; every arm of every experiment replays the same stored detections (718,786 boxes over the split; mAP 0.5613, NDS 0.6480).
Indoor. The ScanNet200 [25] validation split, all 312 scenes, in a streaming instance-segmentation pipeline: class-agnostic 3D instance-mask proposals from Mask3D [26] are labeled per frame by a 2D open-vocabulary detector [8], so each persistent instance receives a stream of per-frame class labels. The proposal set is likewise frozen across arms. This pipeline shares no detector, modality, or codebase beyond the temporal module with the outdoor one.

4.2. Matched controls

Two budgets are matched — emitted detections and emitted identities (Fig. 1b) — and under each we compare the module against a non-temporal score rule and against the other established temporal rule.
Detection-budget-matched control. A score threshold on the frozen baseline detections is calibrated so the control emits *exactly* as many boxes as the temporal module — 257,259 in the sensor frame (threshold 0.2214), 517,566 in the world frame (0.1272), both with zero count error. The match is exact because the count is a step function of the threshold whose breakpoints are the observed scores: the

matched threshold is the K-th largest observed score, ties resolved upward so the count never exceeds K.

Identity-budget-matched control. Per sequence, the control keeps the K highest-scoring tracks (ranked by mean detection score), K being the number of identities the temporal module emits there; counts match exactly (64,428 tracks sensor, 108,225 world). Five additional arms select the same K tracks *at random* (five seeds), scored on association accuracy, separating score-based selection from selection per se. The two budgets cannot hold at once: at equal track count the control emits fewer boxes than the module (190,708 vs. 360,309 sensor; 608,168 vs. 754,500 world), because score-ranked tracks are shorter than persistence-selected ones.

Confidence-accumulation control. To compare persistence against confidence rather than against a static score, a third control propagates a track score by constant decay and fuses it with each matched detection score, following CBMOT's reported best update rule [1]; its parameters are transcribed from the authors' released implementation, not tuned here, except that the gap tolerance is our associator's rather than theirs, since the frozen association is shared by every arm. A single threshold on the refined score is calibrated to the same budget as the arm it is compared with, by the rule above, so this control is matched under both budgets. Its trained score-regression variant is not reproduced: it is fit on nuScenes and would give the control supervision no other arm receives.

Indoor control. The identity-budget design transfers directly: the control keeps the K highest-scoring instances per scene, where K matches the temporal module's emitted-instance count (10,413 instances in total; the control realizes 10,410). Three additional arms select K instances *at random* per scene (three seeds), to separate the effect of score-based selection from that of selection per se.

4.3. Metrics

Outdoor. Official nuScenes mAP and NDS on the emitted box set; class-agnostic HOTA, AssA, DetA, detection recall (DetRe), and IDF1 via TrackEval [21, 24], with similarity $\mathrm{sim} = \mathrm{clip}(1 - d_{xy}/2.0, 0, 1)$ matching the associator's 2.0 m threshold; and official class-aware AMOTA [6] over the 7 nuScenes tracking classes; the two families are reported side by side.

Indoor. The primary endpoint is the per-scene count of *label switches*: frames in which a persistent instance's emitted class label differs from its label in the previous frame *in which it was emitted*. Counting against the last emission rather than the last frame is what makes the count comparable across arms, since the arms emit on different frames. Prediction stability has precedent as a separate axis in video and 3D detection [30, 37], but that family scores box trajectories against ground-truth tracks, which ScanNet200 does not provide, and none of it scores the *class* assigned to a persistent instance. The secondary endpoint is instance-segmentation AP on the final reconstruction.

4.4. Statistics

For all three main comparisons — detection-budget-matched, identity-budget-matched, and indoor — the arms, the primary endpoint, and the criterion for a negative result were fixed before the run in a pre-registration released with the code. For the outdoor comparisons that criterion was that the association gain hold in *both* association frames with scene-bootstrap intervals excluding zero, and that the dataset-level metrics (mAP, AMOTA) move consistently across the two association frames, whichever direction they move in; indoors, that the per-scene label-switch count fall. The causal-emission ablation (Sec. 5.3) predates that protocol and is what motivated it. All comparisons between the temporal module and a control are paired per scene. We report the per-scene Wilcoxon signed-rank test with the rank-biserial correlation as effect size, and a 95% bootstrap confidence interval on the difference of the aggregate metric obtained by resampling scenes (10,000 resamples, fixed seed 20260718). A difference is described as an improvement only when its interval excludes zero; intervals that span zero are reported as no detectable difference, whatever their sign.

Intervals are available for the class-agnostic metrics, which are defined per scene and aggregate linearly over scenes. AMOTA is not, and neither are the nuScenes devkit metrics mAP and NDS: AMOTA integrates MOTA over recall thresholds estimated on the whole split, and mAP and NDS are likewise computed over the whole split, so a per-scene resample does not yield the same estimator. All three are reported as point estimates, and we claim no interval support for AMOTA anywhere; the evidence we offer for it is that its sign and rough magnitude are stable across four independent detection-budget-matched cells — retrospective and causal emission, each in both association frames — and against both the score-threshold and the confidence-accumulation control. The two budgets cannot hold at once: matching emitted identities does not equalise the identities reaching the class-aware evaluation, which scores a subset of classes, so we report AMOTA only under detection matching.

5. Experiments

We report three outdoor comparisons on nuScenes val and one indoor comparison on ScanNet200 val, each between the confirmation-based module and a control granted the same output budget over shared frozen detections. Unless stated otherwise, outdoor results use retrospective emission, whose latency cost is $N-1$ observations; the corresponding frame-index delay is reported in Sec. 5.3.

5.1. Detection-budget-matched comparison

Table 1 compares confirmation against controls calibrated to the identical box count — 257,259 boxes in the sensor frame and 517,566 in the world frame, from the same frozen detector output.

Association accuracy improves. Against the threshold control AssA rises from 0.2119 to 0.2744 in the sensor frame and from 0.5146 to 0.5293 in the world frame, both intervals excluding zero ($[+0.0471, +0.0785]$ and $[+0.0098, +0.0203]$). The per-scene mean gain, a different estimator from the difference of aggregates, is $+0.0460$; the gain holds in 113 of 150 scenes (Wilcoxon p=2.6×10^{-15}, rank-biserial 0.74).

Class-aware tracking accuracy rises. AMOTA goes from 0.0942 to 0.1338 (sensor) and 0.3137 to 0.3864 (world), the largest relative effect we measure. AMOTA admits no per-scene resampling (Sec. 4.4), so this is a point estimate that we rely on only because it moves the same way in both frames and under the comparisons below.

Detection-oriented metrics fall. At the same budget, mAP drops 0.5504 $\rightarrow$ 0.3009 (sensor) and 0.5598 $\rightarrow$ 0.4921 (world), DetA drops 0.3570 $\rightarrow$ 0.2232 ($[-0.1408, -0.1267]$) and 0.2133 $\rightarrow$ 0.1973 ($[-0.0183, -0.0138]$), and detection recall drops 0.6160 $\rightarrow$ 0.4287 ($[-0.2087, -0.1665]$) and 0.6550 $\rightarrow$ 0.6141 ($[-0.0477, -0.0350]$). The DetA loss dominates the geometric mean, so aggregate HOTA falls as well ($[-0.0323, -0.0231]$ sensor; $[-0.0100, -0.0063]$ world).

IDF1 shows no reliable sensor-frame difference. The sensor-frame interval spans zero ($[-0.0108, +0.0037]$) while the paired per-scene test reaches significance (p=7.3×10^{-3}), so the two estimators disagree and we claim no difference; in the world frame IDF1 is slightly *lower* for the confirmation module ($[-0.0135, -0.0070]$).

Against confidence accumulation, the two rules split the metrics. This control isolates *which* temporal rule is used rather than whether one is used at all. Confirmation attains higher AssA in both frames (0.2315 $\rightarrow$ 0.2744 sensor, 0.5110 $\rightarrow$ 0.5293 world; intervals $[+0.0337, +0.0524]$ and $[+0.0146, +0.0225]$), and accumulation is ahead on every other measured axis, also with intervals excluding zero: HOTA, DetA, IDF1 and DetRe (Sec. E). The split extends to detection quality: accumulation leads on mAP (0.4783 $\rightarrow$ 0.3009 sensor, 0.5568 $\rightarrow$ 0.4921 world) and NDS (0.6134 $\rightarrow$ 0.5229, 0.6467 $\rightarrow$ 0.6200).

Scope of the accumulation control. All statements about it concern our reproduction under the configuration of Sec. 4.2, not the original system or its published results (Sec. E).

5.2. Identity-budget-matched comparison

Matching emitted identities rather than boxes (Tab. 2) tests whether the association gain merely reflects having fewer identities to confuse. It does not. AssA remains higher for the confirmation module in both frames (0.2346 $\rightarrow$ 0.2535 sensor, 0.4173 $\rightarrow$ 0.4314 world), with intervals excluding zero ($[+0.0129, +0.0258]$ and $[+0.0101, +0.0185]$) and per-scene wins in 117/150 and 111/150 scenes (p=2.2×10^{-16} and 4.1×10^{-11}). Selecting the same number of tracks *at random* rather than by score reproduces none of it: the seeded arms reach at most 0.2460 (sensor) and 0.3934 (world), both below the confirmation module.

The rule separation survives identity matching. Repeating the comparison against confidence accumulation with *identities* rather than boxes held fixed — the accumulation control ranked by its own refined score and cut at the confirmation module's per-scene identity count, matched exactly — leaves the association separation unchanged: $+0.0252$ sensor ($[+0.0200, +0.0316]$) and $+0.0172$ world ($[+0.0089, +0.0161]$), against $+0.0253$ and $+0.0170$ at matched boxes, with the accompanying losses on HOTA, DetA, IDF1 and DetRe keeping their sign and significance. The separation is the same size whichever currency is held fixed, which excludes the simplest reading of it — that persistence gains association accuracy by carrying fewer identities. The result is robust to the control's ranking and matching granularity (Sec. B).

Budget asymmetry. Identity matching necessarily leaves unequal box counts, because score-ranked tracks are shorter than persistence-selected ones. This biases volume-sensitive metrics against the control — so the AssA result here is conservative — and we therefore read DetA, HOTA and IDF1 from Tab. 1, where boxes are matched exactly.

5.3. What the association gain is bought with

Sections 5.1 and 5.2 price confirmation in detection retention. This section asks what *else* it pays. All results so far release a confirmed track's withheld prefix retrospectively; we first measure the causal alternative, then a policy that avoids both costs.

Causal emission loses the prefix, and with it the world-frame gain. Table 3 repeats the detection-budget-matched comparison with the prefix permanently dropped. The result splits by association frame. The AMOTA gain survives in both (0.0574 $\rightarrow$ 0.0800 sensor, 0.1825 $\rightarrow$ 0.2089 world), and in the sensor frame the association gain is not merely preserved but larger than under retrospective emission (0.2105 $\rightarrow$ 0.2861; interval $[+0.0577, +0.0922]$ against $[+0.0370, +0.0615]$ retrospectively, 110 of 150 scenes). In the world frame, however, causal emission *loses* association accuracy relative to the control (0.4315 $\rightarrow$ 0.4199), with a bootstrap interval entirely below zero ($[-0.0183, -0.0047]$; paired p=0.008, rank-biserial -0.25, confirmation ahead in only 53 of 150 scenes). The detection cost is larger throughout than under retrospective emission (mAP 0.3213 $\rightarrow$

Frame	Arm	Boxes	mAP	NDS	HOTA	AssA	DetA	DetRe	IDF1	AMOTA
Sensor	Baseline (unfiltered)	718,786	0.5613	0.6480	0.1793	0.2007	0.1620	0.6690	0.1112	0.0858
	Control (threshold)	257,259	**0.5504**	**0.6450**	**0.2735**	0.2119	**0.3570**	**0.6160**	0.2082	0.0942
	Control (accumulation)	257,259	0.4783	0.6134	0.2604	0.2315	0.2954	0.5350	0.2146	0.1101
	Confirmation (N=3)	257,259	0.3009	0.5229	0.2462	**0.2744**	0.2232	0.4287	0.2054	**0.1338**
World	Baseline (unfiltered)	718,786	0.5613	0.6480	0.2837	0.5014	0.1613	0.6663	0.2715	0.2896
	Control (threshold)	517,566	0.5598	0.6479	0.3305	0.5146	0.2133	0.6550	0.3391	0.3137
	Control (accumulation)	517,565	0.5568	0.6467	0.3282	0.5110	0.2118	0.6511	0.3399	0.2931
	Confirmation (N=3)	517,566	0.4921	0.6200	0.3224	**0.5293**	0.1973	0.6141	0.3288	**0.3864**

Table 1. **Detection-budget-matched comparison** on nuScenes val (150 scenes, frozen CenterPoint detections, retrospective emission). The three matched arms — a static score threshold (0.2214 sensor, 0.1272 world), a confidence-accumulation control in the manner of [1], and confirmation — emit the same number of boxes. Bold marks the best matched arm, omitted where the leading interval spans zero; the unfiltered baseline is reference only and is not budget-matched.

Arm	Boxes	HOTA	AssA	DetA	IDF1
Sensor frame					
Control (top-K)	190,708	**0.2461**	0.2346	**0.2607**	**0.2081**
Control (rand.)	—	—	.2321–.2460	—	—
Confirmation	360,309	0.1960	**0.2535**	0.1537	0.1531
World frame					
Control (top-K)	608,168	**0.2538**	0.4173	**0.1553**	**0.2596**
Control (rand.)	—	—	.3878–.3934	—	—
Confirmation	754,500	0.2263	**0.4314**	0.1196	0.2121

Table 2. **Identity-budget-matched comparison.** The control keeps, per scene, the K highest-scoring tracks, K being the number of identities the confirmation module emits (64,428 sensor, 108,225 world; exact in both frames); the random-K rows ("rand.") give the range over five seeds of the same selection made at random, scored on AssA only. The box budgets are unequal here, in the control's disfavour on volume-sensitive metrics. AMOTA is omitted: identity matching does not equalise class-aware evaluation volume (Sec. 4.4).

Arm	Boxes	mAP	HOTA	AssA	AMOTA
Sensor frame					
Control (threshold)	221,297	**0.3213**	**0.2417**	0.2105	0.0574
Confirmation, causal	221,297	0.1360	0.2056	**0.2861**	**0.0800**
World frame					
Control (threshold)	503,619	**0.3367**	**0.2729**	**0.4315**	0.1825
Confirmation, causal	503,619	0.2569	0.2456	0.4199	**0.2089**

Table 3. **Causal emission is not free either: detection-budget-matched** (221,297 boxes sensor at threshold 0.2519, 503,619 world at 0.1539; both arms exact). The causal arm drops the pre-confirmation prefix instead of releasing it retrospectively. Intervals are reported for AssA only, the metric the comparison turns on; the remaining columns are point estimates.

0.1360 sensor).

Causal emission is worse where association was already easy: ego-motion compensation leaves the world-frame control associating well on its own, and there the discarded prefix costs more than confirmation cleans up.

How much latency, in which unit. The $N-1$ figure routinely quoted for retrospective emission is correct only in observations (Sec. 3.3): over the 150 scenes the confirmation *frame* horizon has median 3 and 95th percentile 7 — 1.5 s and 3.5 s at the 2 Hz annotated-frame rate, against a constant 2-observation horizon. It is unbounded in principle (Sec. C).

Provisional emission retains the prefix at zero latency. Table 4 measures the proposed policy against the prefix-deleting N=3 confirmation arm over the identical frozen stream. At a retraction horizon of H=4 frames the policy emits 529,662 boxes against that arm's 368,724: it retains 160,938 observations (+43.7%) that the arm deletes, and it does so at an emission latency of exactly zero frames. Detection quality on the emitted set rises accordingly (mAP $0.4483 \to 0.4927$, NDS $0.6013 \to 0.6206$) and so does association consistency (AssA $0.5265 \to 0.5378$).

And it is not free. Over the same comparison the policy gives up class-agnostic detection and identity quality: DetA $0.2275 \to 0.1936$, aggregate HOTA $0.3452 \to 0.3219$, IDF1 $0.3662 \to 0.3305$, and AMOTA before revision falls $0.4051 \to 0.3823$. Emitting a provisional record means emitting records that causal confirmation would have suppressed and that ground truth does not support, and the class-agnostic metrics price exactly that.

Revision recovers the class-aware endpoint. Applying the completed-evidence label to the already-emitted stream takes AMOTA $0.3823 \to 0.4646$ at H=4 (+0.0823), leaving the net effect against the prefix-deleting confirmation arm at +0.0594: revision restores, and then exceeds, an AMOTA the emission step had lost rather than adding one on top of it. It says nothing about association — the class-agnostic matcher does not read the class label, so those columns are a mechanical identity, not a null result.

The retraction horizon saturates. H prices how long a provisional record may still be withdrawn. At H=4 the emitted output becomes identical — byte-identical, verified by checksum — to the unbounded H=∞ policy that defers

Arm	Boxes	mAP	NDS	HOTA	AssA	DetA	IDF1	AMOTA
Baseline (ungated)	718,786	0.5613	0.6480	0.2837	0.5014	0.1613	0.2715	0.2896
Confirmation N=3 (prefix-deleting)	368,724	0.4483	0.6013	0.3452	0.5265	0.2275	0.3662	0.4051
Ours, provisional emission, retraction horizon H=0								
provisional	295,403	0.3832	0.5651	**0.3675**	**0.5619**	**0.2417**	**0.4198**	0.3603
+ revision	295,403	0.3950	0.5701	—	—	—	—	0.4200
Ours, provisional emission, retraction horizon H=4								
provisional	529,662	0.4927	0.6206	0.3219	0.5378	0.1936	0.3305	0.3823
+ revision	529,662	**0.5025**	**0.6247**	—	—	—	—	**0.4646**

Table 4. **Provisional emission and revision**, world frame, nuScenes val (150 scenes, N=3, same frozen detections and association as Tab. 1). These arms are **not budget-matched**: the box count *is* what the emission policy acts on. Revision changes no box, score, or identity, so the class-agnostic metrics are *identical* by construction ("—"). Bold marks the best policy/confirmation value.

the decision to scene end, because no track that ever confirms has a pre-confirmation gap wider than four frames. Every endpoint at $H \geq 4$ therefore coincides by output equality rather than by a tuned metric match. The bound replaces the scene-end decision at no cost in output (Sec. A).

The horizon is itself a trade-off. Tightening it to H=0 — retract at the first missed frame — reverses the picture: the policy then emits fewer boxes than the confirmation arm (295,403) and is the *best* arm in the table on every class-agnostic metric (AssA 0.5619, HOTA 0.3675, DetA 0.2417, IDF1 0.4198), while giving up mAP and AMOTA. We report both endpoints rather than selecting the one that flatters a chosen column: H buys retention and detection quality with class-agnostic identity quality, and H=4 is the setting at which the unbounded-horizon result is recovered in full at bounded latency.

5.4. Confirmation-window sensitivity

All results so far fix N=3. Figure 2 varies it over $\{2, 3, 4, 5\}$, giving each N its own exactly matched control, and asks what the choice is worth. N=1 is excluded: it reproduces the unfiltered baseline detection-for-detection, so its matched control is the same box set and every difference is zero by construction.

The association gain does not depend on the operating point. It is positive at every N in both association frames, and all eight intervals exclude zero. It also grows monotonically with N, close to linearly in the sensor frame and with visible deceleration in the world frame, while AMOTA flattens in the sensor frame and turns over in the world frame, peaking at N=4 (Tab. S1). The detection cost, meanwhile, rises without turning over at any N we measured, and the latency rises linearly by construction.

No single N in this range is best on every metric, and N=3 is not the best on any of them. We report this rather than adjust the operating point: re-selecting N from the sweep would convert an inherited constant into a value tuned on the test metrics, the practice the matched controls exist to guard against. We keep N=3 because N is a latency budget, not a tuned hyperparameter: it is fixed by the $N-1$ observations of delay a deployment can absorb. What the sweep establishes is that the consistency gain is a property of confirmation over the whole range tested, not an artifact of one setting, and that larger windows buy more of it at proportionally more detection cost and latency. The widening gap is driven by the confirmed arm rather than by a failing control, whose own AssA also rises as the matched budget tightens (Tab. S1).

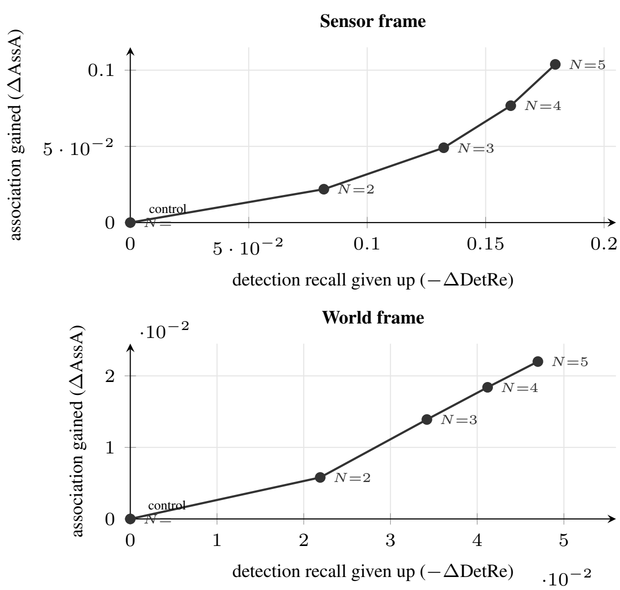

Figure 2. **Confirmation is a trade-off curve, not a tuned setting.** Each point is confirmation at that N minus *its own* detection-budget-matched threshold control, so the control sits at the origin. Moving right along the curve spends detection recall and one further observation of latency, and buys association accuracy. The association gain is positive at every N in both frames and all eight bootstrap intervals exclude zero.

Arm	Inst.	Switches	AP	AP_{50}
Baseline (unfiltered)	—	—	0.1956	—
Control (top-K)	10,410	23,303	0.1926	0.2504
Control (random K)	—	23,275–23,288	—	—
Confirmation (N=3)	10,413	**17,045**	0.1954	0.2532

Table 5. **Indoor comparison** on ScanNet200 val (312 scenes, frozen Mask3D proposals with per-frame open-vocabulary labels). The primary endpoint is the per-scene label-switch count (lower is better, bold); the control keeps the K highest-scoring instances per scene, K matched to the confirmation module's emitted-instance count, and the random-K row spans three seeds (23,275, 23,277, 23,288). The AP columns are secondary and untested. The unfiltered baseline is a validity anchor rather than an arm, and only its AP was recorded, hence its empty cells.

5.5. Indoor: semantic stability

The indoor experiment asks whether the same rule improves *semantic* stability in the pipeline of Sec. 4.1, which shares nothing with the outdoor one. Against the identity-matched control, confirmation reduces label switches from 23,303 to 17,045, a 26.9% reduction at an essentially equal instance budget (10,413 vs. 10,410). The reduction is 18.0 switches at the median (Sec. D) and holds in 309 of 312 scenes (Wilcoxon $p{=}1.5{\times}10^{-52}$, rank-biserial 0.99). The three random-K arms are indistinguishable from the score-ranked control and far from the confirmation module, so the reduction is specific to temporal selection rather than to emitting fewer instances by any rule.

The secondary endpoint shows no material change: AP 0.1954 against 0.1956 unfiltered and 0.1926 for the control. Unlike outdoors, the indoor gain therefore does not come at a measurable detection-quality cost — the two pipelines differ in whether the withheld detections carry segmentation mass, and we report the difference in effect without claiming the outdoor trade-off is avoidable in general.

6. Discussion

6.1. Confirmation is a trade-off, not a free gain

Against an unfiltered baseline, temporal confirmation appears to improve nearly every tracking metric. Against a non-temporal rule granted the same output budget, most of that apparent improvement is the effect of emitting less, and what survives is both sharper and smaller: association consistency improves, and detection-oriented accuracy degrades over an *identical* box set. The gain is bought rather than added, and each additional observation of delay buys more of it at proportionally more detection cost. Against confidence accumulation the picture is a split rather than an ordering — persistence takes association consistency, accumulation keeps the detection- and identity-oriented measures — so the choice between them is a choice of which axis a system is consumed along, not a ranking.

6.2. Provisional emission changes the currency of the cost

Confirmation must decide what to do with the observations that precede it, and the classical rule offers only two answers: release them late, or drop them and lose the world-frame association gain with them. Neither is compatible with a consumer that must act immediately on output whose reliability improves later. The policy's content is a separation — *when output becomes available* is decoupled from *when evidence becomes reliable* — which converts the currency of the cost from latency or retention into revisability: the consumer must accept that a record it holds may be updated or withdrawn, and the retraction horizon bounds how long that obligation lasts. It is an operating-point change, not a new confirmation mechanism.

6.3. Limitations

The revision endpoint is single. AMOTA is the only class-aware metric available to us; the class-agnostic family cannot respond to a semantic-only change, and its invariance in Tab. 4 is mechanical rather than a null result. Whether revision affects identity consistency would need a class-aware association metric, which we do not have and do not claim. Nor do we separate the label change from the nuScenes attribute recomputation it entails, so the AMOTA gain is attributed to the pair.
Scope. We do not measure what a downstream planner or mapper pays to apply a revision or a retraction. The full tracking evaluation is nuScenes only, over a single frozen CenterPoint detection set, so we make no cross-detector claim; ScanNet200 has no instance-tracking ground truth, so the indoor endpoint is label switching rather than an identity metric and tests whether the confirmation rule transfers, not whether the streaming claim holds there.

7. Conclusion

Temporal confirmation provides a genuine association-consistency gain under matched detection and identity budgets, but the gain is coupled to detection retention and to latency. Provisional emission separates immediate output from evidence maturity: observations are emitted causally, then revised or retracted as evidence accumulates. On nuScenes, $H{=}4$ retains 43.7% more observations than the prefix-deleting confirmation arm and revision raises AMOTA by 0.0823 over the provisional output. Against the prefix-deleting arm the policy gives up class-agnostic detection and identity quality. On ScanNet200 the same rule reduces class-label switching by 26.9% at a matched identity count. Confirmation is therefore best understood as an operating point in a detection–association–latency–revisability trade-off rather than a uniformly better tracker.

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A. Retraction horizon

At $H{=}4$ a track is confirmed or retracted within five frames (2.5 s) of its last observation, instead of at scene end (up to 40 frames, 20 s). The bound costs nothing in output — the emitted set is byte-identical to $H{=}\infty$ — and downstream event traffic is unchanged by it (161.60 events per frame at every H).

B. Control ranking robustness

Ranking the identity-budget control by mean instead of peak refined score, or matching identities dataset-wide instead of per scene, moves its AssA by less than 0.004 in both association frames.

C. Confirmation latency distribution

The $N-1$ figure quoted for retrospective emission is a count of *observations*. Over the 150 scenes the confirmation *frame* horizon has mean 3.29, median 3, 95th percentile 7 and maximum 10; at the 2 Hz annotated-frame rate this is a median of 1.5 s with a tail of 5.0 s, against a constant 2-observation horizon. It is unbounded in principle.

D. Indoor per-scene distribution

The per-scene label-switch reduction is 18.0 switches at the median and 20.1 on average (95% CI on the mean $[18.7, 21.5]$).

E. Confidence-accumulation control

At the matched detection budget the accumulation control leads confirmation on HOTA ($[-0.0157, -0.0128]$ sensor, $[-0.0065, -0.0052]$ world), DetA ($[-0.0805, -0.0645]$, $[-0.0162, -0.0131]$), IDF1 ($[-0.0115, -0.0071]$, $[-0.0123, -0.0099]$) and DetRe; all intervals exclude zero. In the world frame the control is not statistically separable from the static threshold on any of the five metrics we resample (all intervals span zero); in the sensor frame it is.

Frame	N	Delay	Boxes	ΔAssA	95% CI	ΔAMOTA	ΔDetRe
Sensor	2	1	551,452	$+0.0219$	$[+0.0155, +0.0286]$	$+0.0177$	-0.0817
	3	2	360,309	$+0.0491$	$[+0.0370, +0.0615]$	$+0.0339$	-0.1323
	4	3	269,330	$+0.0767$	$[+0.0596, +0.0936]$	$+0.0393$	-0.1606
	5	4	217,295	$+0.1038$	$[+0.0820, +0.1249]$	$+0.0414$	-0.1794
World	2	1	869,354	$+0.0058$	$[+0.0021, +0.0096]$	$+0.0228$	-0.0219
	3	2	754,500	$+0.0139$	$[+0.0086, +0.0195]$	$+0.0378$	-0.0342
	4	3	658,099	$+0.0184$	$[+0.0120, +0.0252]$	$+0.0424$	-0.0412
	5	4	576,711	$+0.0220$	$[+0.0148, +0.0297]$	$+0.0381$	-0.0470

Table S1. **Confirmation-window sensitivity.** Each row is confirmation at that N with retrospective emission, minus *its own* detection-budget-matched threshold control, matched exactly to the box count in the Boxes column. Delay is $N-1$ observations. The $N=3$ rows are matched at a larger budget than Tab. 1 and are not the same comparison. ΔAMOTA is a whole-split point estimate; the DetRe intervals are omitted for space and exclude zero at every N. The matched control's own AssA rises with N as its budget tightens, from 0.1980 at $N=2$ to 0.2107 at $N=5$ in the sensor frame and from 0.4131 to 0.4262 in the world frame. This sweep is an exploratory sensitivity analysis.